

## SOA Survey

# Semantic Contextual Reasoning for Object Separability in Aerial Object Detection

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## Abstract

Aerial object detection from satellite and drone imagery is a critical capability for remote sensing, urban planning, and surveillance applications. However, detecting objects in overhead views poses unique challenges including extreme scale variation, dense object arrangements, arbitrary orientations, and visually cluttered backgrounds that cause detectors to confuse foreground objects with their surroundings. This survey reviews eight key papers spanning foundational detection architectures (Faster R-CNN, YOLO, FPN, RetinaNet), aerial-specific methods for oriented and small object detection (SCRDet, RoI Transformer), and a recent transformer-based detector tailored for UAV imagery (HEDS-DETR), supplemented by an applied thesis on drone-based detection. Our review identifies a persistent gap: while prior work addresses scale, rotation, and density individually, no existing method explicitly models semantic contextual reasoning to improve object separability from visually similar backgrounds in dense aerial scenes. We propose to investigate a context-aware feature enhancement module integrated into an existing detection framework and evaluated on the DOTA dataset to address this gap.

## 1. Problem Definition

Object detection in aerial imagery involves localizing and classifying objects of interest—such as vehicles, ships, aircraft, and buildings—within overhead images captured by satellites or unmanned aerial vehicles (UAVs). Unlike natural images taken from a ground-level perspective, aerial images present a bird’s-eye view where objects appear at vastly different scales, can be oriented in any direction, and are frequently packed together in dense clusters. This task is fundamental to applications in remote sensing, traffic monitoring, disaster response, and precision agriculture.

The problem is particularly challenging from a deep learning standpoint for several interrelated reasons. First, many objects of interest occupy only a small fraction of the high-resolution input image, making small object detection extremely difficult as feature representations become impoverished through successive downsampling layers. Second, objects in aerial scenes can appear at arbitrary angles, which breaks the horizontal bounding box assumption baked into most standard detectors. Third, dense arrangements of similar-looking objects (e.g., rows of parked cars, docked ships) cause severe confusion during both region proposal and classification stages. Standard non-maximum suppression (NMS) can erroneously eliminate correct detections in such scenarios.

Most critically, aerial images contain visually complex backgrounds—roads, rooftops, vegetation, water—that share texture and color patterns with the objects of interest. Current detection pipelines lack an explicit mechanism for semantic contextual reasoning that would allow the network to leverage surrounding scene context to distinguish a vehicle from a similarly textured road patch, or a ship from the harbor structure beneath it. This background confusion problem is the central difficulty that motivates our survey and proposed direction.

## 2. Literature Review

We organize our reading around four core challenges that the aerial object detection community has been working to address. Each subsection discusses the relevant papers, their approaches, and their remaining shortcomings.

## 2.1. Challenge 1: Multi-Scale Feature Representation

Detecting objects across extreme scale ranges is a foundational problem in object detection. The Feature Pyramid Network (FPN) [3] introduced a top-down architecture with lateral connections that builds semantically rich feature maps at all scales from a single input image. By merging high-resolution, spatially precise low-level features with semantically strong high-level features, FPN dramatically improved detection of small objects. When integrated into a basic Faster R-CNN system [1], FPN achieved state-of-the-art results on COCO with an 8-point improvement in average recall for small objects.

However, FPN alone does not fully solve the small object problem in aerial imagery, where objects can be far smaller relative to the image than in natural scenes. SCRDet [5] addresses this with its Sampling Fusion Network (SF-Net), which fuses C3 and C4 features from ResNet with a flexible anchor stride, allowing finer sampling that better captures small objects. Their experiments on DOTA showed that reducing the anchor stride from 16 to 6 improved oriented bounding box mAP by over 3 points. More recently, HEDS-DETR [7] tackles this in a transformer-based framework by introducing an Efficient Small Object Pyramid (ESOP) that reintegrates high-resolution S2 features discarded by the baseline RT-DETR, yielding measurable gains on the VisDrone dataset. Despite these advances, multi-scale feature fusion strategies still struggle when small objects are embedded within visually similar background textures, since the features themselves may not be discriminative enough regardless of resolution.

## 2.2. Challenge 2: Oriented and Dense Object Detection

Objects in aerial images appear at arbitrary orientations, making standard axis-aligned bounding boxes inadequate. Horizontal bounding boxes around rotated objects include substantial background, leading to feature contamination and poor classification. The RoI Transformer [6] addresses this by learning to convert horizontal region proposals into rotated ones through a lightweight fully connected layer, then extracting rotation-invariant features via

rotated position-sensitive RoI alignment. This approach achieved 69.56% mAP on DOTA with FPN, significantly outperforming horizontal-box baselines.

SCRDet [5] tackles orientation from the loss function side, introducing an IoU-Smooth L1 loss that adds an IoU constant factor to handle the boundary discontinuity problem in angle regression. This eliminates the sudden loss spikes that occur when an object's angle is near the boundary of the defined range. Faster R-CNN [1] introduced the anchor-based region proposal paradigm that underlies most of these methods, using multi-scale, multi-aspect-ratio anchors to cover diverse object shapes. However, extending this to rotated anchors—as attempted by methods like RRPN—dramatically increases the number of anchors and computational cost.

For dense detection, the fundamental tension is between NMS (which suppresses overlapping detections) and recall (which requires retaining them). RetinaNet [4] demonstrated that the foreground-background class imbalance in dense one-stage detectors could be addressed through focal loss, which down-weights well-classified examples to focus training on hard cases. This insight is broadly relevant to aerial detection, where background examples vastly outnumber foreground ones. HEDS-DETR [7] sidesteps NMS entirely through its end-to-end DETR architecture, but compensates for decoder instability via Selective Query Recollection (SQR) to prevent cascading prediction errors across decoder stages.

## 2.3. Challenge 3: Background Noise and Feature Discrimination

The most directly relevant challenge to our proposed direction is the difficulty of separating objects from visually similar backgrounds. In aerial imagery, the background is not merely a uniform canvas but a complex scene with textures, structures, and patterns that can closely resemble the objects of interest. SCRDet [5] addresses this with its supervised Multi-Dimensional Attention Network (MDA-Net), which combines pixel-level and channel-level attention to suppress background noise and highlight foreground features. Crucially, the pixel attention is supervised using binary maps derived from ground-truth annotations, which guides the network to learn a meaningful

foreground-background separation rather than discovering it in an unsupervised manner. This approach improved mAP by 3.67% on DOTA over the baseline.

HEDS-DETR [7] proposes the High-Frequency Enhanced Semantics Network (HFESNet) backbone, which preserves high-frequency details (edges, textures) alongside deep semantic features. The rationale is that conventional backbones suppress the fine-grained signals needed to resolve overlapping objects. Their Geometry-Aware Position Encoding (GAPE) further injects explicit spatial priors into the decoder’s attention mechanism, improving localization for tightly clustered objects.

YOLO [2] introduced the idea that seeing the entire image at once provides implicit contextual reasoning, since each prediction is conditioned on global information. This reduces background false positives compared to sliding-window and region-proposal methods that only see local patches. However, YOLO’s contextual reasoning is implicit and coarse-grained; it does not model semantic relationships between an object and its immediate surroundings in a structured way.

The thesis by Karsan [8] provides practical insights into deploying YOLO-based detectors on edge devices for drone imagery, highlighting that accuracy degradation in real-world aerial scenes is strongly tied to the inability to distinguish objects from cluttered backgrounds, even when the detector is otherwise well-optimized for speed.

While attention mechanisms (SCRDet) and frequency-aware backbones (HEDS-DETR) improve feature quality, they do not explicitly model semantic context—the idea that an object’s identity can be disambiguated by reasoning about what surrounds it. This remains an open gap.

#### 2.4. Challenge 4: Computational Efficiency and Real-Time Constraints

Aerial detection systems often operate under tight latency requirements, particularly on UAV platforms with limited compute. YOLO [2] pioneered the single-shot paradigm that frames detection as a regression problem, achieving 45 FPS on a Titan X GPU. RetinaNet [4] demonstrated that one-stage detectors could match two-stage accuracy with focal loss while main-

taining reasonable speeds. HEDS-DETR [7] achieves 131.6 FPS on an RTX 3090 while reducing parameters by 17% compared to its RT-DETR baseline, showing that architectural enhancements for aerial detection need not come at prohibitive cost.

However, adding context-aware modules inevitably introduces computational overhead. Any proposed context enhancement must be lightweight enough to maintain real-time performance, which constrains the design space for our proposed direction.

#### Open Problems (flagged by the field):

- Explicit semantic context modeling for object-background separability in cluttered aerial scenes remains largely unexplored [5, 6].
- Generalization of aerial detectors across different sensor types, altitudes, and geographic regions is poorly studied [8].
- Data efficiency remains underexplored; most aerial detection models require large-scale annotated datasets that are costly to obtain, limiting scalability across domains [8, 7].
- Weakly supervised aerial detection using partial or coarse labels (e.g., point-level or image-level annotations) is not well understood, particularly in dense scenes where precise localization is critical [5, 6].

### 3. Research Gap and Proposed Direction

The literature review reveals that significant progress has been made on individual challenges—multi-scale features (FPN, SF-Net, ESOP), oriented detection (RoI Transformer, IoU-Smooth L1), dense detection (focal loss, SQR), and feature discrimination (MDA-Net, HFESNet). However, the problem of semantic contextual reasoning for object separability remains inadequately addressed. Current attention-based approaches like MDA-Net improve feature quality by suppressing noise, but they operate on individual feature maps without explicitly reasoning about the semantic relationship between an object region and its surrounding context. Similarly, frequency-aware backbones preserve fine-grained details but do not structure this information around

object-context boundaries.

We identified three potential gaps from our reading: (1) semantic context modeling for object-background separation, (2) data efficiency in aerial object detection, where most models require large-scale annotated datasets that are costly to obtain and limit scalability across domains, and (3) weakly supervised aerial detection using partial or coarse labels (e.g., point-level or image-level annotations), which is not well understood particularly in dense scenes where precise localization is critical. We select the first gap because it is both more significant—background confusion is consistently identified as a primary error source across the surveyed papers—and more tractable within a semester project, since it can be implemented as a modular feature enhancement block added to an existing pretrained detector. The remaining two gaps, while important, require fundamentally different experimental setups involving annotation-scarce regimes and novel supervision paradigms that are harder to scope within our timeline and compute constraints.

Prior work in aerial object detection struggles with distinguishing objects from visually similar backgrounds in dense scenes. We propose a context-aware feature enhancement module integrated into a detection framework, evaluated on the DOTA dataset.

**Research Question:** Can a lightweight semantic context aggregation module, inserted into a pretrained detection backbone, measurably improve object-background separability and detection accuracy for dense, small objects in aerial imagery, as measured by mean average precision on a subset of the DOTA benchmark?

#### 4. Benchmark, Evaluation, and Data Landscape

**Datasets.** Our primary evaluation dataset is DOTA, which is the largest and most widely used benchmark for oriented object detection in aerial images. It is freely available and provides both horizontal and oriented bounding box annotations. For a feasible semester project, we plan to use a subset of 3–5 object classes and resized image patches (e.g.,  $512 \times 512$ ) to manage computational cost.

Table 1: Key datasets in the field.

| Dataset        | Size          | Modality      | Notes                                                         |
|----------------|---------------|---------------|---------------------------------------------------------------|
| DOTA           | 2,806<br>imgs | RGB<br>Aerial | 15 classes,<br>OBB+HBB<br>annotations,<br>188K in-<br>stances |
| NWPU<br>VHR-10 | 800<br>imgs   | RGB<br>VHR    | 10 classes,<br>from Google<br>Earth                           |
| VisDrone       | 8,629<br>imgs | RGB<br>Drone  | 10 classes,<br>up to<br>$2000 \times 1500$<br>px              |
| HRSC2016       | 1,061<br>imgs | RGB<br>Aerial | Ship detec-<br>tion, OBB                                      |

**Evaluation Protocol.** The community primarily uses mean Average Precision (mAP) at IoU threshold 0.5, computed separately for horizontal bounding box (HBB) and oriented bounding box (OBB) tasks. DOTA provides an official evaluation server for standardized comparison. mAP is the appropriate metric because it captures both precision and recall across confidence thresholds, and per-class AP breakdowns reveal whether improvements are concentrated on specific object types or are general. For our context module evaluation, we will also report per-class AP to identify whether gains are strongest for categories most affected by background confusion (e.g., small vehicles, ships in harbors).

**Compute Context.** The surveyed papers used hardware ranging from NVIDIA GTX 1080 (8GB) for SCRDet to dual RTX 8000 GPUs for HEDS-DETR. Most experiments on DOTA were conducted on single GPUs with 8–12GB VRAM. Our project will use Google Colab or Kaggle (free tier), with COCO-pretrained weights for initialization, training on a DOTA subset for 20–50 epochs with image resizing to  $512 \times 512$  to keep training time within a few hours per experiment.

## Experimental Design Considerations

- **Baselines:** Our primary baseline would be a pre-trained Faster R-CNN or YOLOv5 model fine-tuned on the same DOTA subset without the proposed context module. We would also compare against the same backbone with a standard SE-Net channel attention block (as used in SCRDet’s MDA-Net) to isolate the contribution of semantic context reasoning beyond simple attention.
- **Ablation design:** We would isolate (a) the context aggregation module itself (with/without), (b) the spatial extent of context used (local neighborhood vs. larger receptive field), and (c) the supervision signal for the context module (unsupervised attention vs. ground-truth-guided). These three variables directly correspond to the design choices that distinguish our approach from existing attention methods.
- **Evaluation pitfalls:** The DOTA dataset has known evaluation concerns. Multi-scale testing with image pyramids can inflate results significantly (SCRDet gained nearly 3% mAP from image pyramids alone), so we must report single-scale results for fair comparison. Additionally, class-specific NMS thresholds can be tuned to game per-class AP; we should use uniform thresholds. The large image sizes in DOTA require patch-based inference with overlap stitching, and inconsistent patch sizes between methods can affect comparability.

## 5. Annotated Reading List

### References

- [1] S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *NeurIPS*, 2015. *What it does:* Introduces Region Proposal Networks (RPNs) that share convolutional features with the detection network, enabling nearly cost-free region proposals. *What it contributes:* Unifies region proposal generation and object detection into a single end-to-end trainable network, achieving 5–17 FPS with state-of-the-art accuracy on PASCAL VOC. *Why it is here:* Faster R-CNN is the foundational two-stage detection framework upon which most aerial detection methods, including our potential baselines, are built.
- [2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi. You Only Look Once: Unified, Real-Time Object Detection. *CVPR*, 2016. *What it does:* Frames object detection as a single regression problem from image pixels to bounding box coordinates and class probabilities, processed in one network evaluation. *What it contributes:* Demonstrates that a unified single-shot architecture can achieve real-time detection (45 FPS) with strong generalization, though at the cost of localization precision for small objects. *Why it is here:* YOLO establishes the single-shot detection paradigm and provides implicit global context reasoning that motivates our interest in explicit context modeling.
- [3] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie. Feature Pyramid Networks for Object Detection. *CVPR*, 2017. *What it does:* Builds a multi-scale feature pyramid with strong semantics at all levels using a top-down pathway with lateral connections from a standard ConvNet backbone. *What it contributes:* Provides a general-purpose feature extraction framework that significantly improves detection of small objects by enriching high-resolution features with high-level semantics. *Why it is here:* FPN is the backbone architecture used by most aerial detectors in our survey and is the multi-scale foundation upon which our context module would be integrated.
- [4] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal Loss for Dense Object Detection. *ICCV*, 2017. *What it does:* Identifies foreground-background class imbalance as the primary obstacle for one-stage detectors and proposes focal loss to down-weight well-classified examples during training. *What it contributes:* Enables one-stage detectors (RetinaNet) to surpass two-stage detector accuracy for the first time, achieving 39.1 AP on COCO with a simple FPN-based architecture. *Why it is here:* Focal loss directly addresses the class imbalance problem that is amplified in aerial

*scenes, and informs our understanding of how loss design affects detection in dense settings.*

- [5] X. Yang, J. Yang, J. Yan, Y. Zhang, T. Zhang, Z. Guo, X. Sun, and K. Fu. SCRDet: Towards More Robust Detection for Small, Cluttered and Rotated Objects. ICCV, 2019. *What it does: Proposes a multi-category rotation detector combining a sampling fusion network, supervised multi-dimensional attention, and an IoU-aware rotation loss for aerial images. What it contributes: Achieves 72.61% mAP on DOTA by jointly addressing small object detection (SF-Net), background noise suppression (MDA-Net), and rotation boundary problems (IoU-Smooth L1 loss). Why it is here: SCRDet’s MDA-Net is the closest existing work to our proposed context module, and understanding its supervised pixel/channel attention design is essential for positioning our contribution.*
- [6] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu. Learning RoI Transformer for Oriented Object Detection in Aerial Images. CVPR, 2019. *What it does: Proposes a learnable RoI Transformer that converts horizontal region proposals into rotated ones, extracting rotation-invariant features via rotated position-sensitive RoI alignment. What it contributes: Achieves state-of-the-art 69.56% mAP on DOTA without needing rotated anchors, while adding negligible computational overhead to a Light-Head R-CNN baseline. Why it is here: RoI Transformer demonstrates that orientation can be handled as a learned spatial transformation rather than anchor engineering, informing the design space for our context module.*
- [7] H. Peng, L. Chen, H. Zhu, and Y. Chen. High-Frequency Semantics and Geometric Priors for End-to-End Detection Transformers in Challenging UAV Imagery. *arXiv:2507.00825v3*, 2025. *What it does: Enhances RT-DETR for aerial detection with a frequency-aware backbone (HFESNet), an efficient small object pyramid (ESOP), geometry-aware position encoding (GAPE), and selective query recollection (SQR). What it contributes: Achieves +3.8% AP and +5.1% AP50 improvement over RT-DETR-*
- R18 on VisDrone while reducing parameters by 17% and maintaining real-time performance at 131.6 FPS. Why it is here: Represents the latest transformer-based approach to aerial detection and demonstrates that preserving high-frequency details and providing geometric priors are complementary to—but distinct from—semantic context reasoning.*
- [8] N. Karsan. Performance Analysis and Evaluation of Object Detection Algorithms for Drone Networks. *Master Thesis, University of Padova*, 2024. *What it does: Evaluates YOLOv5 and YOLOv8 on the Stanford Drone and VisDrone datasets with edge-device optimization (OpenVINO, NCNN) on Raspberry Pi 4. What it contributes: Provides practical benchmarks on accuracy-efficiency trade-offs for drone-deployed detectors, showing that optimization frameworks can maintain reasonable accuracy under severe compute constraints. Why it is here: Offers real-world deployment perspective on aerial detection limitations, confirming that background confusion remains a primary error source even in well-optimized systems.*

## 6. Preliminary Implementation Resources

- <https://github.com/ultralytics/yolov5> Official YOLOv5 repository; candidate baseline detector for fine-tuning on DOTA subset with our proposed context module.
- <https://github.com/DetectionTeamUCAS> SCRDet and related aerial detection implementations (TensorFlow); reference for MDA-Net attention module design and DOTA evaluation pipeline.
- <https://github.com/open-mmlab/mmdetection> OpenMMLab toolbox for rotated object detection; provides standardized implementations of RoI Transformer, rotated Faster R-CNN, and DOTA data loading/evaluation utilities.
- <https://github.com/open-mmlab/mmdetection> General object detection toolbox; includes Faster R-CNN, FPN, and RetinaNet implementations that serve as potential baseline

frameworks.

- [https://github.com/CAPTAIN-WHU/DOTA\\_devkit](https://github.com/CAPTAIN-WHU/DOTA_devkit) Official DOTA dataset development kit; essential for image splitting, result merging, and evaluation submission.
- <https://github.com/lyuwenyu/RT-DETR> Official RT-DETR implementation; baseline for the HEDS-DETR approach, useful as a reference for transformer-based aerial detection.
- [https://github.com/hukaixuan19970627/YOLOv5\\_DOTA\\_OBB](https://github.com/hukaixuan19970627/YOLOv5_DOTA_OBB) YOLOv5 adapted for oriented bounding box detection on DOTA; directly relevant as a lightweight baseline for our experiments.
- <https://github.com/jbwang1997/OBBDetection> Oriented bounding box detection toolbox built on MMDetection; provides data augmentation and evaluation tools specific to aerial datasets.